

Mahika Nair

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EDUCATION

- MSc Data Science | University of Edinburgh, United Kingdom** *Sep 2025 - Aug 2026*
- Completed August 2026 (On track for Distinction, awaiting final award)
 - Current Average: 86.6%
- PG Diploma in AI-ML and Leadership | Plaksha University, India** *Aug 2023 - Jul 2024*
- CGPA: 9.86/10.0
- BSc (Hons.) Mathematics | Krea University, India** *Aug 2020 - Jul 2023*
- Minor in Chemistry
 - CGPA: 9.24/10.0

WORK EXPERIENCE

- Predli Consulting AB | Associate AI Engineer** *Sep 2024 - Aug 2025*
- Built an AI-driven document processing and RAG pipeline (LLMs, OCR, vector retrieval) that cut manual processing time from days to minutes (95% reduction).
 - Designed agentic AI workflows (AutoGen) for the ResolveWithAI platform, improving response accuracy from 80% to 90% and growing adoption to 10+ organic users within 3 months.
 - Built full-stack features (TypeScript) and LangGraph-based agent orchestration for Predli Studio, integrating MCP-compatible tools for autonomous multi-step workflows.
- Max Super Specialty Hospital | Summer Research Intern** *Jul 2023 - Sep 2023*
- Developed MRI-based tumour segmentation workflows and extracted quantitative radiomics features using PyRadiomics for breast cancer analysis.
 - Evaluated radiomic feature importance to identify imaging characteristics associated with tumour heterogeneity and clinical outcomes.
- Fractal Analytics | Imagineer Intern** *Jun 2022 - Aug 2022*
- Mined and cleaned client sales data to build revenue growth models, supporting data-driven decision-making through structured analysis of large datasets.
- Krea University | Summer Research Intern** *Jun 2021 - Aug 2021*
- Conducted computational analysis of Ubiquitin and SUMO protein families using structural bioinformatics and molecular visualisation tools (Chimera) to investigate evolutionary and structural relationships and their influence on protein function.

PROJECTS

- ML Surrogate Modelling for Protein Adsorption | Master's Dissertation** *May 2026 - Aug 2026*
- Developed the first deep learning surrogate for Self-Consistent Field Theory (SCFT) protein adsorption simulations, predicting residue-level equilibrium distances with 0.4 nm MAE and achieving a 200× speedup.

- Built a hybrid Transformer and U-Net architecture in PyTorch, integrating coarse-grained sequences, ESM protein language model embeddings, and SCFT targets to learn sequence-dependent adsorption behaviour.
- Established model robustness through ablation, out-of-distribution, and interaction-strength analyses, using GPU-accelerated HPC workflows with Linux and Slurm.

Satellite Crop Classification | Capstone Thesis, Plaksha University

Apr 2024 - Jun 2024

- Collaborated with Professor Shashank Tamaskar to classify crop fields in Punjab using high-resolution Sentinel-2 satellite imagery and Google Earth Engine, providing government stakeholders with timely insights for resource allocation.
- Pretrained a pixel-based vision transformer (PRESTO) on high-resolution Indian satellite data, achieving an F1-score of 85.5% for accurate crop classification.

'The Office' RAG Chatbot | Personal Project

Feb 2024 - Mar 2024

- Developed a RAG-based chatbot using Streamlit, LangChain, Chroma DB, Gemini, and YouTube API, combining Wikipedia embeddings with relevant video links to answer fan queries with fast semantic retrieval.

Mudra Visualisation with CV | Personal Project

Jan 2024 - Feb 2024

- Built a YOLOv8 model detecting 30 Indian classical hand gestures in real time (92% accuracy), trained on a self-created dataset of 200 images per class sourced from 35+ individuals.

PUBLICATIONS

- Piyush Sharma, Sneha Thomas, Mahika Nair, and Ananth Govind Rajan, "Machine Learnable and Rapidly Interpretable Language for Nanopores Enables Structure-Property Relationships in Nanoporous 2D Materials", *Journal of the American Chemical Society* (2024) [[DOI](#)]

SKILLS

- **Programming Languages:** Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV), R, SQL, TypeScript
- **ML/AI:** Transformers, Hugging Face, LLMs, NLP, Computer Vision, RAG, LangChain, LangGraph, AutoGen, MCP, Multi-Agent Systems
- **Infrastructure:** AWS (EC2, ECS, SageMaker), Docker, Linux, Slurm, Git, REST APIs

AWARDS

Plaksha University

2023 - 2024

- Academic Excellence Award for securing a top 2 CGPA in the cohort
- Selected as the Graduation Speaker for the cohort
- 1st place in the startup pitching competition of the Entrepreneurial Challenge Lab

Krea University

2020 - 2023

- 1st Year Dean's List student